

Vertical Deposition Profile of Cleanser Spray

Braeden Scott ^a,

^a Aerosol Research and Engineering Laboratories Inc., Overland Park, Kansas – 66213

Study Info:

ARE Labs Project # 11048.10

Date: January 29, 2026

Study Sponsor:

Bragg & Bevier LLC

Testing Lab:

ARE Labs Inc.
12880 Metcalf Avenue
Overland Park, KS – 66213

Purpose: To characterize vertical spray reach and internal surface coverage of a single-use cleanser spray when deployed inside a closed 96-gallon residential trash bin using gravimetric mass-gain deposition coupons (mg/in²).

Background: In enclosed volumes, internal deposition is governed by spray plume geometry, dispersion, wall impaction, gravitational settling, and post-deposition migration (dripping/pooling). Products marketed as “foaming” may exhibit foam-rise effects; however, when discharge behavior is spray-dominant, coverage is primarily a function of spray transport and settling dynamics.

Methods: Five aerosol cans were evaluated using one full, continuous upright discharge per can (five independent trials). Deposition was quantified using 1 in² absorbent filter paper coupons installed at four locations: 1 ft, 2 ft, and 3 ft above the bin bottom on the interior wall and the interior lid surface. Four coupons were installed at each location (16 coupons per trial). Testing was performed with the bin lid closed during discharge and a 3-minute dwell period at controlled conditions of 70°F and 60% RH. Coupons were pre-weighed and post-weighed on an analytical balance and deposition density was reported as mg/in². Internal lighting and video documentation were used to support qualitative interpretation.

Results: Trial 1 was impacted by an unexpected fan-shaped spray and can tip-over, resulting in highly directional impingement, incomplete measurements, and only one coupon showing measurable deposition (106 mg/in²). Trial 2 produced a complete dataset (n = 16), with mean deposition of 12.00 mg/in² (1 ft), 5.75 mg/in² (2 ft), 156.75 mg/in² (3 ft), and 23.00 mg/in² (lid), characterized by non-uniform deposition with multiple 0 mg/in² coupons and isolated high-mass events. Trials 3 and 4, using modified nozzles, discharged primarily as a coherent stream with a consistent directional bias, yielding substantially lower deposition (Trial 3: 5.75, 0.75, 8.50, 0.50 mg/in²; Trial 4: 0.75, 0.50, 0.75, 0.75 mg/in² at 1 ft, 2 ft, 3 ft, and lid, respectively). Trial 5 produced moderate deposition (11.25, 1.75, 10.00, and 26.00 mg/in²), with high variability at the 3 ft and lid locations (±1 SD up to 50.02 mg/in²), indicating localized impaction rather than uniform coverage.

Conclusion: Across the completed datasets, the cleanser spray deposited measurable mass on elevated internal surfaces; however, deposition was highly variable and strongly influenced by discharge behavior and plume geometry. Trial 2 showed substantial deposition at the 3 ft wall and measurable lid deposition, occurring as localized impaction rather than uniform coverage. Trials 3 and 4 exhibited stream-like discharge behavior, resulting in minimal deposition at all locations and preferential wetting of one side of the bin. Trial 5 produced moderate deposition at 1 ft and 3 ft with the highest mean deposition on the lid, but remained highly non-uniform. Trial 1 is retained for traceability only due to documented deviations.

Study Overview

The objective of this non-GLP screening study was to quantify internal coverage efficiency and vertical reach of a single-use cleanser spray in a closed 96-gallon bin using area-normalized gravimetric deposition coupons. Five independent full-discharge trials were planned and executed as one discharge per can. Deposition was assessed at four locations spanning the internal vertical profile and lid interior using sixteen 1 in² coupons per trial to generate an engineering-relevant deposition profile for enclosed-bin application performance.

Background

Closed-bin deposition is driven by the interaction of spray plume geometry and momentum with the enclosure boundaries and by subsequent transport mechanisms, including gravitational settling, turbulent dispersion, impaction, and post-deposition migration via dripping and pooling. These mechanisms can produce strong spatial heterogeneity even when total discharged mass is high. When product behavior is spray-dominant rather than foam-dominant, “vertical reach” is best interpreted as the ability of the spray plume and entrained droplets to reach elevated surfaces (upper walls and lid) rather than foam-rise height.

Test Configuration

Testing was performed in a closed 96-gallon residential trash bin configured to represent enclosed, in-use deployment conditions. The spray can was positioned upright at the center of the bin floor and actuated as a single continuous full discharge while the lid remained closed. Deposition was measured using 1 in × 1 in (1.0 in²) absorbent filter paper coupons mounted flush to the interior surfaces at four locations: 1 ft, 2 ft, and 3 ft above the bin bottom along the interior wall and on the interior lid surface. Four coupons were installed per location (16 total per trial) with wall coupons distributed circumferentially and lid coupons distributed across representative lid regions. The bin was equipped with internal LED illumination and an internally mounted, sealed smartphone camera for video documentation. Testing was conducted under controlled laboratory conditions of 70°F and 60% RH, and coupons were collected after a 3-minute closed-lid dwell period for gravimetric determination of deposition density (mg/in²).

Deviations / Trial 1 Deviation

Trial 1 included a major deviation identified during execution. The nozzle/orifice configuration was different than expected and produced a fan-shaped spray pattern rather than a cone spray. This altered plume geometry increased directional impingement and reduced circumferential distribution, and several coupon post-mass measurements as well as the surface deposition were not recorded. As a result, Trial 1 is documented for traceability but is not used as the primary dataset for coverage efficiency interpretation. **(To see data go to Table A1 in the appendix).**

Testing Conditions

Testing was performed at ARE Labs under controlled environmental conditions of 70°F and 60% RH. A single 96-gallon residential trash bin was used as the test enclosure, and the lid remained closed during discharge and the post-discharge dwell period.

Samples Tested

The test articles consisted of pressurized, single-use cleanser spray cans in which the nozzle locks itself once activated. The spray cans are designed to dispense a cleaning formulation as a spray plume. Each can was discharged in a single continuous actuation with the can maintained upright. During testing, the product exhibited

spray-dominant behavior rather than sustained foam formation, and performance was therefore interpreted primarily in terms of spray transport, impaction, and settling within the enclosed bin.



Figure 1: The spray can samples used for testing.



Figure 2: The nozzle of the spray can samples 1&2.

Major Equipment Utilized

Testing utilized a 96-gallon bin enclosure, internal LED lighting, a smartphone video recording system mounted to the underside of the lid (sealed for protection), and an analytical balance suitable for mg-level measurements (per

ARE Labs equipment inventory). A measuring tool and height markers were used to establish coupon locations at 1 ft, 2 ft, and 3 ft above the bin bottom and on the interior lid surface.



Figure 3: The 96-gallon trash bin used for all test trials.

Methods

Each trial consisted of placing one aerosol can upright at the center of the bin floor, initiating internal video recording, and closing the bin lid. The can was discharged continuously to empty while the lid remained closed. After discharge, the bin remained closed for a 3-minute dwell period to allow spray settling and deposition stabilization. Coupons were then removed sequentially with minimal disturbance, placed into labeled containers, and weighed.

Deposition coupons were 1 in × 1 in absorbent filter paper (1.0 in²). Sixteen coupons were installed per trial, with four coupons at each location: 1 ft, 2 ft, and 3 ft on the interior wall and four coupons on the interior lid surface. Each coupon was uniquely labeled and pre-weighed prior to installation. Post-test mass was recorded, mass gain was calculated as post minus pre, and deposition density was reported directly as mg/in².



Figure 4: Coupon placement and spray configuration inside the 96-gallon bin (1ft, 2ft, 3ft wall locations and interior lid surface).

RESULTS: Trial 1 (Deviation / Partial Data)

Trial 1 was impacted by a fan-shaped spray pattern associated with an unexpected nozzle/orifice configuration as well as the can tipping over. Several coupon post-test masses were not recorded, and only one coupon showed measurable deposition. The one recorded deposition event was 106 mg/in² (Coupon ID 1; 64 mg pre to 170 mg post). The can mass decreased from 131.51 g to 77.29 g, corresponding to an observed discharged mass of 54.22 g. Due to altered plume geometry and incomplete measurements, Trial 1 results are considered potentially skewed and are not used for deposition profiling.

Spray Can Mass			
Trial #	Initial Mass of Spray Can (no lid) (g)	Post-Testing Mass (g)	Spray Formula Mass (g)
Trial #1	131.51	77.29	54.22
Trial #2	156.82	65.18	91.64
Trial #3	149.94	63.98	85.96
Trial #4	162.27	64.58	97.69
Trial #5	135.40	91.48	43.92

Table 1: Spray can mass pre-test and post-test (discharged mass by difference).

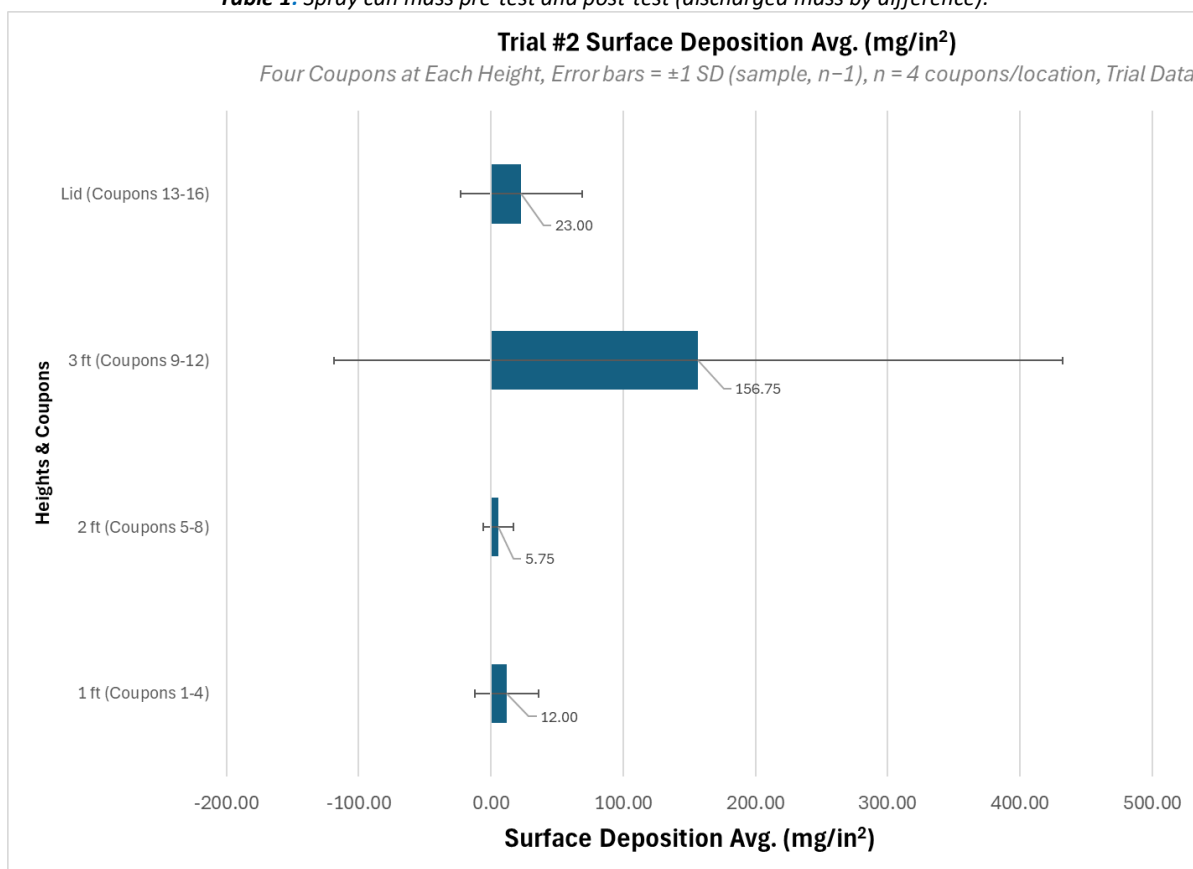


Figure 5: Surface deposition average (mg/in²) by location for Trial #2.

Trial 2 (Primary Dataset)

Trial 2 produced a complete dataset for all sixteen coupons ($n = 4$ per location). Mean deposition by location was 12.00 mg/in² at 1 ft, 5.75 mg/in² at 2 ft, 156.75 mg/in² at 3 ft, and 23.00 mg/in² on the interior lid surface. Deposition was non-uniform at all

locations, with multiple 0 mg/in² coupons and isolated high deposition events driving the location means. Ranges were 0–48 mg/in² (1 ft), 0–23 mg/in² (2 ft), 0–568 mg/in² (3 ft), and 0–92 mg/in² (lid). The can mass decreased from 156.82 g to 65.18 g, corresponding to an observed discharged mass of 91.64 g.

Coupon Mass Average Trial #2

# ID	Initial Mass Avg. (mg)	Post-testing Mass Avg. (mg)	Surface Deposition Avg. (mg/in ²)
1 ft (Coupons 1-4)	64.25	76.25	12.00
2 ft (Coupons 5-8)	65.00	70.75	5.75
3 ft (Coupons 9-12)	63.00	219.75	156.75
Lid (Coupons 13-16)	71.50	94.50	23.00

Table 2: Coupon pre-test, post-test, and surface deposition averages at each measurement level.

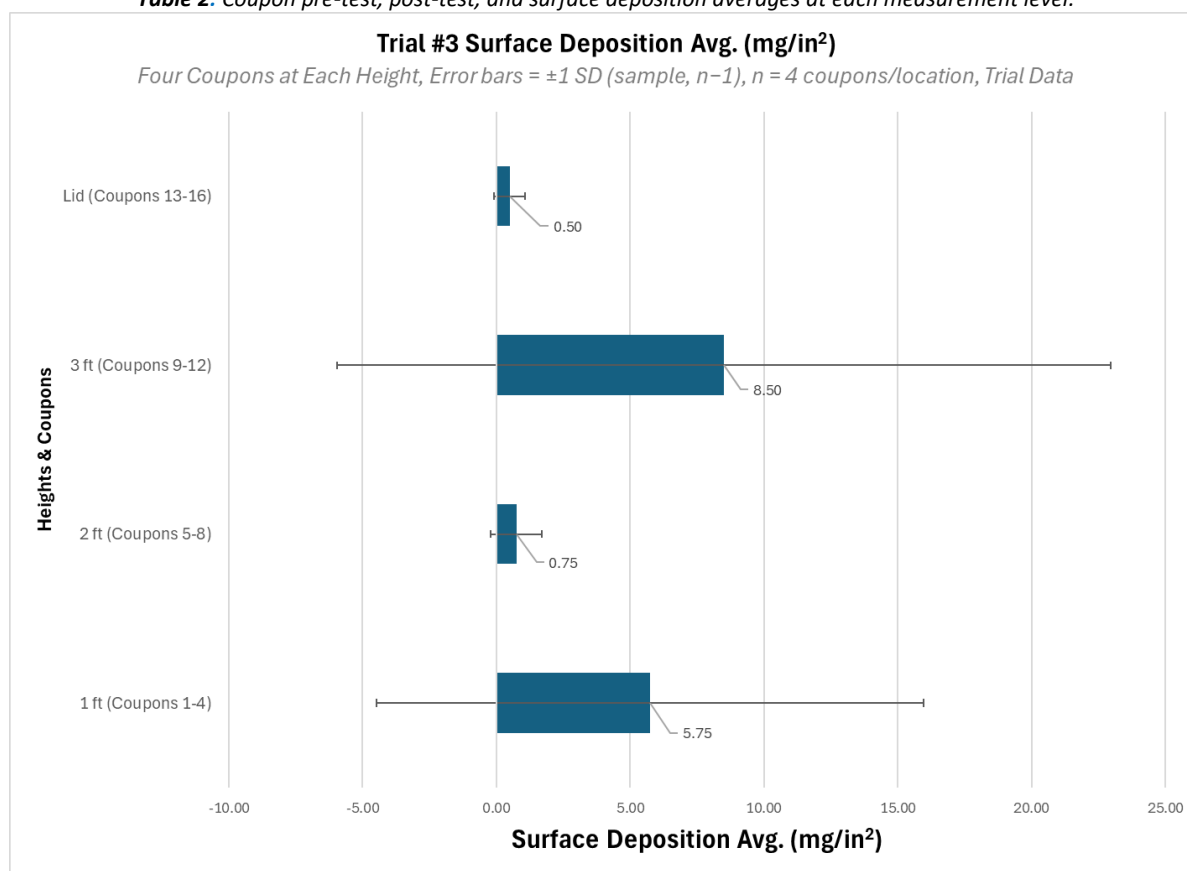


Figure 6: Surface deposition average (mg/ in²) by location for Trial #3.

Trial 3 (Modified Nozzle; Streaming Discharge Observed)

Trial 3 was performed using a newly supplied can with a modified nozzle intended to produce a full cone spray. Qualitatively, the discharge behavior was predominantly a coherent stream for most of the can life, transitioning toward a more dispersed spray

pattern only near end-of-can conditions (approximately the final ~10% remaining). Mean deposition by location ($n = 4$ coupons/location) was 5.75 mg/in² at 1 ft, 0.75 mg/in² at 2 ft, 8.50 mg/in² at 3 ft, and 0.50 mg/in² on the interior lid surface. The observed magnitude and distribution are consistent with limited droplet dispersion and directional impingement during the “streaming” portion of the discharge.

Coupon Mass Average Trial #3

# ID	Initial Mass Avg. (mg)	Post-testing Mass Avg. (mg)	Surface Deposition Avg. (mg/in ²)
1 ft (Coupons 1-4)	62.00	67.75	5.75
2 ft (Coupons 5-8)	68.50	69.25	0.75
3 ft (Coupons 9-12)	73.50	82.00	8.50
Lid (Coupons 13-16)	76.50	77.00	0.50

Table 3: Coupon pre-test, post-test, and surface deposition averages at each measurement level.

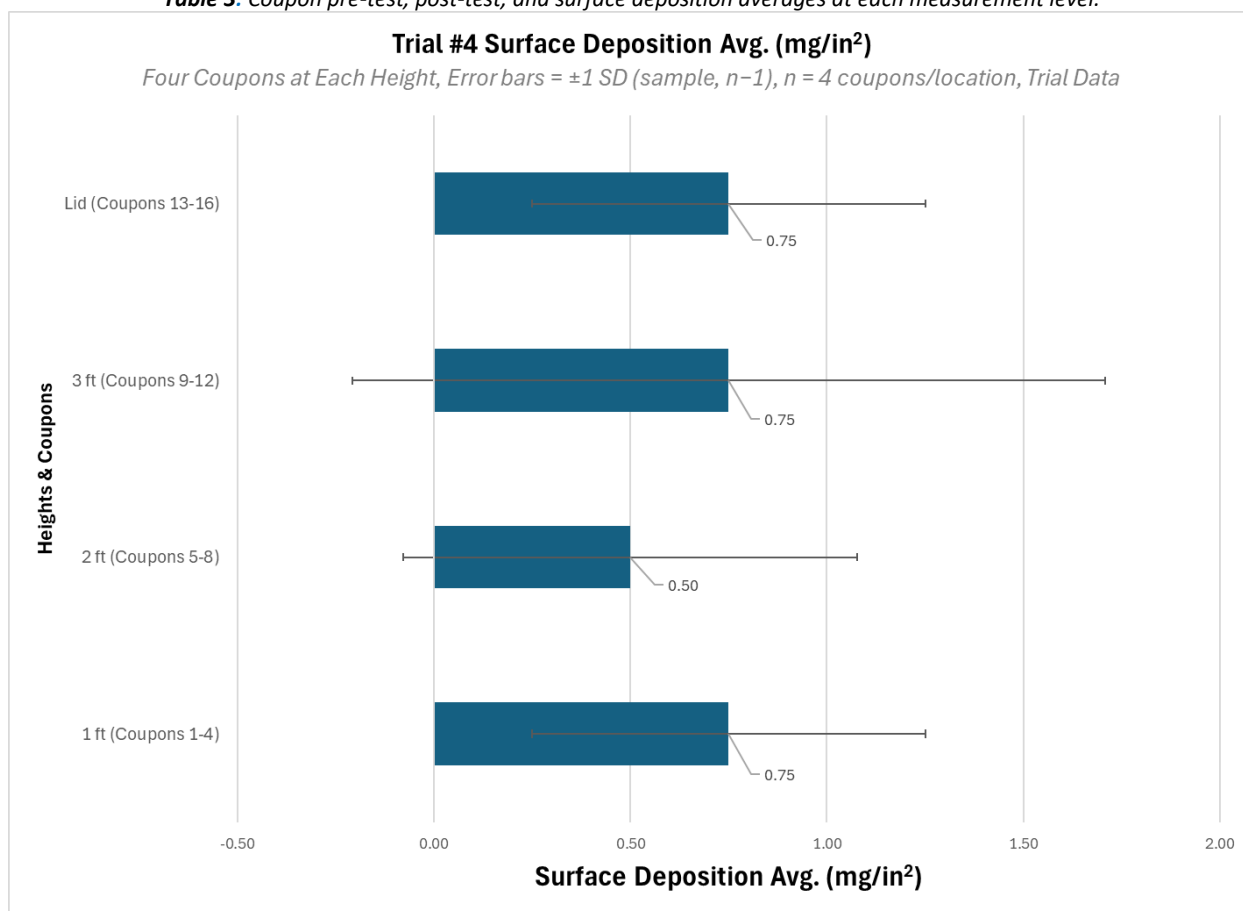


Figure 7: Surface deposition average (mg/in²) by location for Trial #4

Trial 4 (Modified Nozzle; Streaming Discharge Observed)

Trial 4 was performed using a second newly supplied can of the same modified-nozzle configuration. As in Trial 3, the discharge was predominantly a coherent stream through most of the can life, transitioning toward a more dispersed spray pattern only

near end-of-can conditions. Mean deposition by location ($n = 4$ coupons/location) was 0.75 mg/in² at 1 ft, 0.50 mg/in² at 2 ft, 0.75 mg/in² at 3 ft, and 0.75 mg/in² on the interior lid surface. Relative to Trial 2, deposition magnitudes were substantially reduced and more consistent with a low-dispersion discharge regime.

Coupon Mass Average Trial #4

# ID	Initial Mass Avg. (mg)	Post-testing Mass Avg. (mg)	Surface Deposition Avg. (mg/in ²)
1 ft (Coupons 1-4)	72.25	73.00	0.75
2 ft (Coupons 5-8)	66.00	66.50	0.50
3 ft (Coupons 9-12)	68.25	69.00	0.75
Lid (Coupons 13-16)	65.75	66.50	0.75

Table 4: Coupon pre-test, post-test, and surface deposition averages at each measurement level.

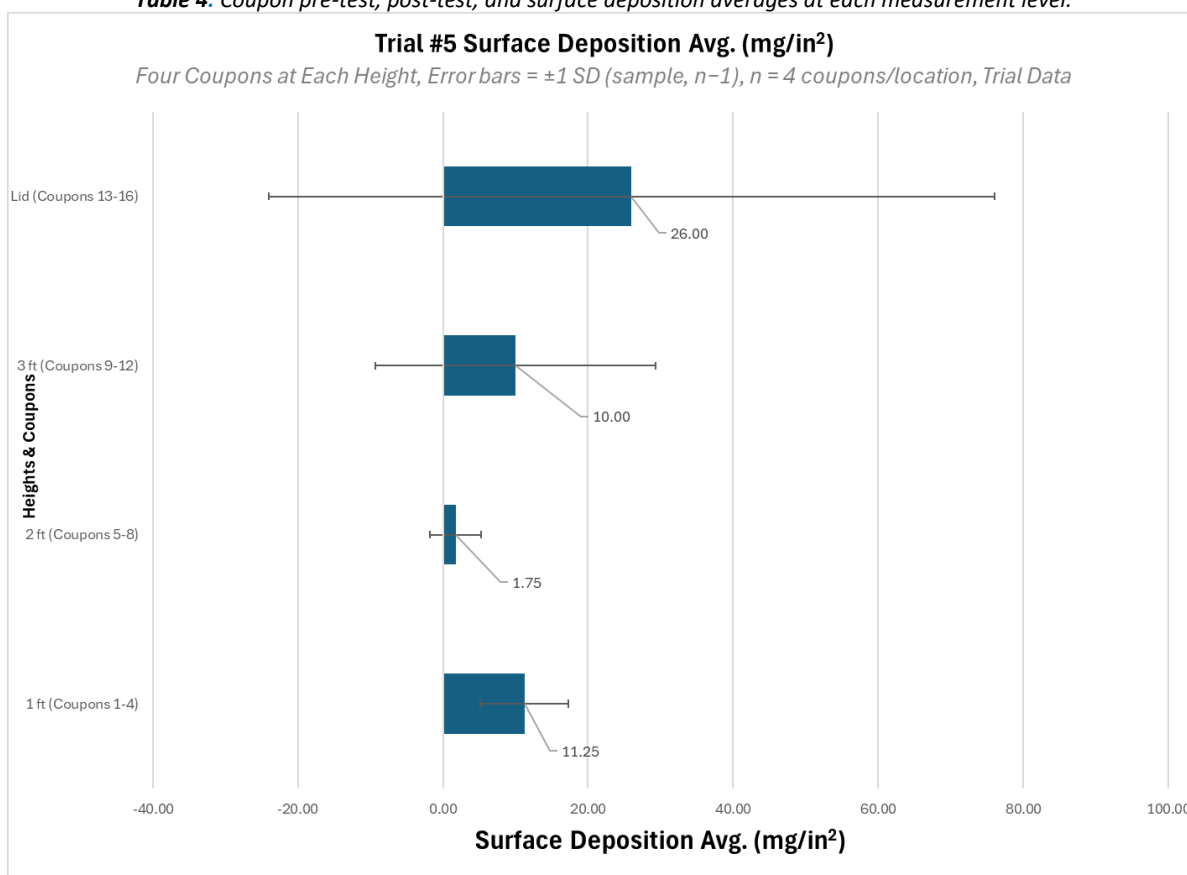


Figure 8: Surface deposition average (mg/ in²) by location for Trial #5.

Trial 5 (Modified Nozzle; Streaming Discharge Observed)

Trial 5 was performed per the same closed-bin discharge protocol and coupon deposition measurement approach ($n = 4$ coupons/location). Mean deposition by location was 11.25 mg/in² at 1 ft, 1.75 mg/in² at 2 ft, 10.00 mg/in² at 3 ft,

and 26.00 mg/in² on the interior lid surface. Deposition variability was highest at the lid and 3 ft locations (± 1 SD, sample $n-1$, of 50.02 mg/in² and 19.34 mg/in², respectively), consistent with non-uniform deposition dominated by localized high-deposition events.

Coupon Mass Average Trial #5

# ID	Initial Mass Avg. (mg)	Post-testing Mass Avg. (mg)	Surface Deposition Avg. (mg/in ²)
1 ft (Coupons 1-4)	66.50	77.75	11.25
2 ft (Coupons 5-8)	65.25	67	1.75
3 ft (Coupons 9-12)	70.25	80.25	10.00
Lid (Coupons 13-16)	65.75	91.75	26.00

Table 5: Coupon pre-test, post-test, and surface deposition averages at each measurement level.

Deviations/Observations

Additional Observations (Trials 3–5): Three new spray cans were supplied with a modified nozzle intended to produce a full cone spray. During testing, discharge behavior for Trials 3–5 was predominantly a coherent stream for the majority of can life, with the intended dispersed spray behavior occurring primarily near end-of-can conditions (~10% remaining). The discharge trajectory was consistently biased in the same direction, resulting in preferential wetting of approximately one side of the bin interior. Subsequent evaluation identified that the actuation latch/trigger mechanism becomes angled after actuation and imposes an off-axis load on the actuator/nozzle during discharge, causing the spray plume to deflect toward the latch direction.

Corrective Actions / Recommendations

To mitigate the consistent off-axis plume bias observed in Trials 3–5, corrective actions should focus on eliminating latch-induced actuator tilt and increasing the sensitivity of the actuator/nozzle to side loading. Recommended actions include valve selection (lower spring rate), smoother latch mechanics, and reduced friction so the user doesn't need to push as hard. Redesigning the latch/trigger to apply actuation force through the actuator centerline (or symmetrically), and adding positive actuator centering/guidance features (e.g., pilot/guide on the valve stem interface) to prevent the actuator from cocking under load. A fixtured actuation test that bypasses the latch is recommended to confirm restoration of coaxial plume alignment prior to additional coverage evaluations.

Conclusions

Across the completed datasets, the cleanser spray demonstrated the ability to deposit measurable mass on elevated interior surfaces, including the interior lid; however, deposition behavior was strongly influenced by discharge behavior and plume geometry during the single-use actuation. When the discharge was characterized by a dispersed spray (Trial 2), substantial deposition was achieved at the upper wall (3 ft) and on the lid, confirming that vertical reach to elevated surfaces is achievable. This deposition, however, occurred as localized high-impaction events rather than uniform circumferential wetting.

In contrast, Trials 3 and 4 exhibited discharge behavior dominated by a coherent stream during most of the actuation, resulting in substantially reduced deposition at all vertical locations and minimal lid coverage. Trial 5 produced intermediate results, with moderate deposition at 1 ft and 3 ft and the highest mean deposition on the lid; however, large standard deviations at the 3 ft and lid locations indicate continued non-uniform deposition driven by isolated impaction and coalescence events.

Overall, the results indicate that vertical deposition performance for this single-actuation fogger is highly sensitive to actuator alignment and plume stability during discharge. The consistent off-axis plume bias observed in Trials 3–5 was attributed to latch/trigger-induced actuator tilt, demonstrating that relatively small mechanical asymmetries during actuation can materially limit coverage uniformity. Trial 1 is retained for traceability only and is not considered representative due to nozzle deviation, can instability, and incomplete measurements.

References

- [1]. Hinds, W.C., *Aerosol Technology: Properties, Behavior, and Measurement of Airborne Particles*, Wiley-Interscience.
- [2]. Baron, P.A., & Willeke, K. (Eds.), *Aerosol Measurements: Principles, Techniques, and Applications*, Wiley.
- [3]. ISO 16711:2015, *Workplace air quality — Gravimetric determination of respirable dust fraction mass concentration*, ISO.
- [1]. ANSI/ASHRAE Standard 145.1-2015, "Laboratory Test Method for Assessing the Performance of Gas-Phase Air-Cleaning Systems: Loose Granular Media."
- [2]. US EPA Compendium Method TO-15 – *Determination of Volatile Organic Compounds (VOCs)*.
- [3]. ARE Labs SOP-FTIR-001: Operation and Calibration of Thermo Max-FTIR Gas Analyzer.

Testing Facility

Aerosol Research and Engineering Labs, Inc.
12880 Metcalf Ave.
Overland Park, KS 66213

Client: Bragg & Bevier LLC

ARE Project: 11048.10

Study Director

Jamie Balarashti
Aerosol Research and Engineering Laboratories

Conflict of Interest

Aerosol Research and Engineering Laboratories, Inc., have no affiliations with, or involvement in any capacity, with Bragg & Bevier LLC's financial interests such as membership, employment, stock ownership, or other equity interests.

Study Coordinator



Weston Schaper
Vice-President, R&D

1/29/2026

Principal Investigator



Braeden Scott
Lab Technician, ARE Lab Inc.

1/29/2026

APPENDIX A

Coupon Mass Data (Trials 1–5)

Coupon Mass Trial #1			
# ID	Initial Mass (mg)	Post-testing Mass (mg)	Mass Gain (mg)
1	64	170	106
2	61	-	N/A
3	64	-	N/A
4	63	-	N/A
5	67	-	N/A
6	68	-	N/A
7	61	-	N/A
8	64	-	N/A
9	66	-	N/A
10	64	-	N/A
11	64	-	N/A
12	58	-	N/A
13	69	-	N/A
14	70	-	N/A
15	73	-	N/A
16	74	-	N/A

Table A1: Coupon mass pre-test, post-test, and mass gain for Trial 1 (partial dataset).

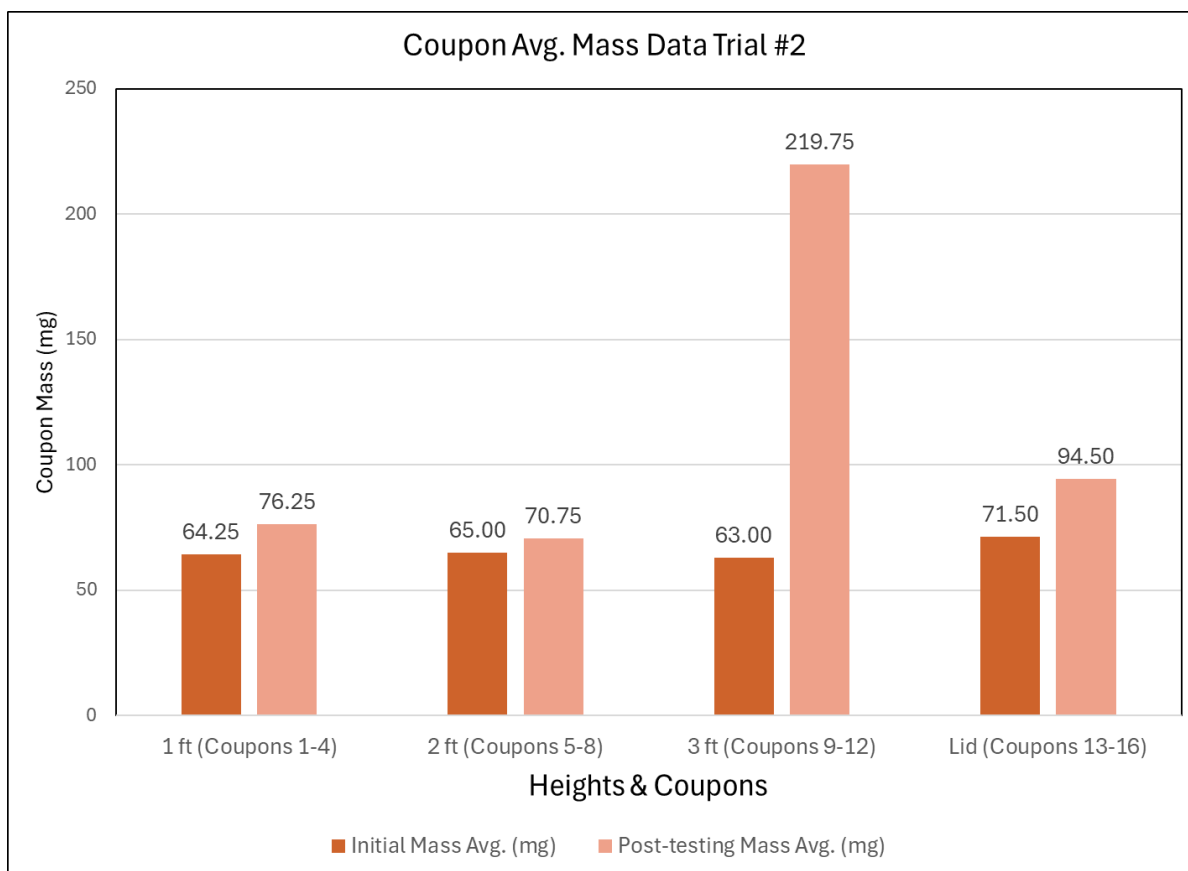


Figure A1: Coupon mass pre-test and post-test (Trial 2).

Coupon Mass Trial #2

# ID	Initial Mass (mg)	Post-testing Mass (mg)	Mass Gain (mg)
1	66	66	0
2	61	61	0
3	67	115	48
4	63	63	0
5	67	67	0
6	68	68	0
7	61	61	0
8	64	87	23
9	66	66	0
10	64	64	0
11	64	632	568
12	58	117	59
13	69	69	0
14	70	162	92
15	73	73	0
16	74	74	0

Table A2: Coupon mass pre-test, post-test, and mass gain for Trial 2.

Coupon Mass Trial #3

# ID	Initial Mass (mg)	Post-testing Mass (mg)	Mass Gain (mg)
1	60	60	0
2	65	86	21
3	63	65	2
4	60	60	0
5	65	66	1
6	69	71	2
7	69	69	0
8	71	71	0
9	69	73	4
10	73	103	30
11	74	74	0
12	78	78	0
13	74	75	1
14	77	77	0
15	78	78	0
16	77	78	1

Table A3: Coupon mass pre-test, post-test, and mass gain for Trial 3.

Coupon Mass Trial #4

# ID	Initial Mass (mg)	Post-testing Mass (mg)	Mass Gain (mg)
1	74	75	1
2	74	75	1
3	74	75	1
4	67	67	0
5	67	68	1
6	64	65	1
7	67	67	0
8	66	66	0
9	67	68	1
10	69	71	2
11	70	70	0
12	67	67	0
13	68	69	1
14	64	65	1
15	65	65	0
16	66	67	1

Table A4: Coupon mass pre-test, post-test, and mass gain for Trial 4.

Coupon Mass Trial #5

# ID	Initial Mass (mg)	Post-testing Mass (mg)	Mass Gain (mg)
1	69	78	9
2	63	77	14
3	67	71	4
4	67	85	18
5	67	67	0
6	62	69	7
7	65	65	0
8	67	67	0
9	69	70	1
10	73	112	39
11	67	67	0
12	72	72	0
13	68	71	3
14	65	65	0
15	64	64	0
16	66	167	101

Table A5: Coupon mass pre-test, post-test, and mass gain for Trial 5.



Figure A2: Samples 3-5 nozzle.